Estimation Theory

Kalman Filter

Discrete-Time Linear System

$$
\boxed{
\begin{aligned}
x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\
y_k &= H_k x_k + v_k
\end{aligned}
}
$$

Measurement noise.

process noise

$$
\begin{aligned}
w_k &\sim (0, Q_k) \\
v_k &\sim (0, R_k) \\
\mathbb{E}(w_k w_j^T) &= Q_k \delta_{k-j} \\
\mathbb{E}(v_k v_j^T) &= R_k \delta_{k-j} \\
\mathbb{E}(v_k w_j^T) &= 0
\end{aligned}
$$

Goal: To estimate x_k based on the noisy measurements $\{y_k\}$.

Notations:

- $\hat{x}_k^+ = \mathbb{E}[x_k | y_1, \ldots, y_k]$ (a posteriori estimate)
- $\hat{x}_k^- = \mathbb{E}[x_k | y_1, \ldots, y_{k-1}]$ (a priori estimate)
- $\hat{x}_0^+ = \mathbb{E}[x_0]$
- $P_k^- = \mathbb{E}[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T]$
- $P_k^+ = \mathbb{E}[(x_k - \hat{x}_k^+)(x_k - \hat{x}_k^+)^T]$

Cov. Matrix

Time k에서의 state를 나타냄

ex) $(\hat{x}_k^+, P_k^+)$

앞 Time의 posterior 현재의 prior.

① Dynamic Update $(\hat{x}_{k-1}^{+} \rightarrow \hat{x}_k^{-},\ P_{k-1}^{+} \rightarrow P_k^{-})$

$$\boxed{\begin{aligned} x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \end{aligned}}$$

Mean of $\omega = 0$

$$\begin{aligned} \hat{x}_k^{-} &= \mathbb{E}[x_k|y_1,\ldots,y_{k-1}] \\ &= \mathbb{E}[F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + \underbrace{w_{k-1}}_{\sim 0}|y_1,\ldots,y_{k-1}] \\ &= F_{k-1}\mathbb{E}[x_{k-1}|y_1,\ldots,y_{k-1}] + G_{k-1}u_{k-1} \\ &= F_{k-1}\hat{x}_{k-1}^{+} + G_{k-1}u_{k-1} \end{aligned}$$

Repeating previous term

$$\begin{aligned} P_k^{-} &= \mathbb{E}[(x_k - \hat{x}_k^{-})(x_k - \hat{x}_k^{-})^T] \\ &= \mathbb{E}[(F_{k-1}x_{k-1} + \cancel{G_{k-1}u_{k-1}} + w_{k-1} - (F_{k-1}\hat{x}_{k-1}^{+} + \cancel{G_{k-1}u_{k-1}}))(\ \cdot\ )^T] \\ &= \mathbb{E}[(F_{k-1}(x_{k-1} - \hat{x}_{k-1}^{+}) + w_{k-1})(\ \cdot\ )^T] \\ &= F_{k-1}\mathbb{E}[(x_{k-1} - \hat{x}_{k-1}^{+})(x_{k-1} - \hat{x}_{k-1}^{+})^T]F_{k-1}^T + \mathbb{E}[w_{k-1}w_{k-1}^T] \\ &= F_{k-1}P_{k-1}^{+}F_{k-1}^T + Q_{k-1} \end{aligned}$$

2 Measurement Update $(\hat{x}_k^- \to \hat{x}_k^+,\ P_k^- \to P_k^+)$

Since we have

$y_k = H_k x_k + v_k$.

- $\hat{x}_k^- := \mathbb{E}[x_k | y_1, \ldots, y_{k-1}]$
- $\hat{y}_k := \mathbb{E}[y_k | y_1, \ldots, y_{k-1}] = H_k \hat{x}_k^-$

Cov. Matrix for v_k.

- $C_{yy} := \mathbb{E}[(y_k - \hat{y}_k)(y_k - \hat{y}_k)^T | \cdot] = \mathbb{E}[(H_k(x_k - \hat{x}_k^-) + v_k)(\cdot)^T | \cdot] = H_k P_k^- H_k^T + R_k$
- $C_{xy} := \mathbb{E}[(x_k - \hat{x}_k)(y_k - \hat{y}_k)^T | \cdot] = \mathbb{E}[(x_k - \hat{x}_k^-)(H_k(x_k - \hat{x}_k^-) + v_k)^T | \cdot] = P_k^- H_k^T,$

LMMSE estimator is:

- $\underline{\hat{x}_k^+} = \hat{x}_k^- + C_{xy} C_{yy}^{-1} (y_k - \hat{y}_k) = \underline{\hat{x}_k^- + P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} (y_k - H_k \hat{x}_k^-)}$
- $\underline{P_k^+} = \underline{P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-}$

Let $K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}$. Then

K_k : Kalman Gain

- $\hat{x}_k^+ = \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-)$
- $P_k^+ = P_k^- - K_k H_k P_k^-$

Some Equalities

$$P_k^+ = P_k^- - K_k H_k P_k^-$$
$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \rightsquigarrow K_k(HP^-H^T+R) = P^-H^T$$

$$\begin{aligned}
P_k^+ &= P_k^- - K_k H_k P_k^- = (I - K_k H_k) P_k^- \\
&= P_k^- - K_k H_k P_k^- - P_k^- H_k^T K_k^T + P_k^- H_k^T K_k^T \\
&= P_k^- - P_k^- H_k^T K_k^T - K_k H_k P_k^- + K_k (H_k P_k^- H_k^T + R_k) K_k^T \\
&= (P_k^- - P_k^- H_k^T K_k^T - K_k H_k P_k^- + K_k H_k P_k^- H_k^T K_k^T) + K_k R_k K_k^T \\
&= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T
\end{aligned}$$

$$\begin{aligned}
P_k^+ &= P_k^- - K_k H_k P_k^- \\
&= P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- \\
&= ((P_k^-)^{-1} + H_k^T R_k^{-1} H_k)^{-1} \qquad \text{(matrix inversion lemma)}
\end{aligned}$$

$$\begin{aligned}
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R^k)^{-1} \\
&= \boxed{P_k^+ (P_k^+)^{-1}} P_k^- H_k^T (H_k P_k^- H_k^T + R^k)^{-1} \\
I \quad &= P_k^+ ((P_k^-)^{-1} + H_k^T R_k^{-1} H_k) P_k^- H_k^T (H_k P_k^- H_k^T + R^k)^{-1} \\
&= (P_k^+ H_k^T + P_k^+ H_k^T R_k^{-1} H_k P_k^- H_k^T)(H_k P_k^- H_k^T + R^k)^{-1} \\
&= P_k^+ H_k^T (I + R_k^{-1} H_k P_k^- H_k^T)(H_k P_k^- H_k^T + R^k)^{-1} \\
&= P_k^+ H_k^T R_k^{-1} (R_k + H_k P_k^- H_k^T)(H_k P_k^- H_k^T + R^k)^{-1} \\
&= P_k^+ H_k^T R_k^{-1}
\end{aligned}$$

Discrete-Time Kalman Filter

1. The dynamic system is given by the following equations:

$$
\begin{aligned}
x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\
y_k &= H_k x_k + v_k \\
E(w_k w_j^T) &= Q_k \delta_{k-j} \\
E(v_k v_j^T) &= R_k \delta_{k-j} \\
E(w_k v_j^T) &= 0
\end{aligned}
\tag{5.17}
$$

2. The Kalman filter is initialized as follows:

$$
\begin{aligned}
\hat{x}_0^+ &= E(x_0) \\
P_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]
\end{aligned}
\tag{5.18}
$$

3. The Kalman filter is given by the following equations, which are computed for each time step $k = 1, 2, \cdots$:

$$
\begin{aligned}
P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
&= P_k^+ H_k^T R_k^{-1} \\
\hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} = \textit{a priori} \text{ state estimate} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k\hat{x}_k^-) = \textit{a posteriori} \text{ state estimate} \\
P_k^+ &= (I - K_k H_k)P_k^-(I - K_k H_k)^T + K_k R_k K_k^T \\
&= [(P_k^-)^{-1} + H_k^T R_k^{-1} H_k]^{-1} \\
&= (I - K_k H_k)P_k^-
\end{aligned}
\tag{5.19}
$$

numerically more stable

Properties of Kalman Filter

Let $\tilde{x}_k = x_k - \hat{x}_k$ be the estimation error and consider the problem of finding an estimator which minimizes (at each time step):

$$\min \mathbb{E}[\tilde{x}_k^T S_k \tilde{x}_k],$$

where S_k is user-defined positive definite matrix.

- If $\{w_k\}$ and $\{v_k\}$ are Gaussian, zero-mean, uncorrelated, and white, then the Kalman filter is the solution to the above problem.
- If $\{w_k\}$ and $\{v_k\}$ are zero-mean, uncorrelated, and white, then the Kalman filter is the optimal *linear* filter. $\because$ Kalman filter = LMMSE

Measurement update equation: $\hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - H_k \hat{x}_k^-)$

- $(y_k - H_k \hat{x}_k^-)$ is called the *innovations* - part of the measurement which contains new information about the state.

One-Step Kalman Filter Equations

- *a priori* Kalman filter

$$\begin{aligned}
\hat{x}_{k+1}^{-} &= F_k \hat{x}_k^{+} + G_k u_k \\
&= F_k[\hat{x}_k^{-} + K_k(y_k - H_k \hat{x}_k^{-})] + G_k u_k \\
&= F_k(I - K_k H_k)\hat{x}_k^{-} + F_k K_k y_k + G_k u_k \\
P_{k+1}^{-} &= F_k P_k^{+} F_k^T + Q_k \\
&= F_k(P_k^{-} - K_k H_k P_k^{-})F_k^T + Q_k \\
&= F_k P_k^{-} F_k^T - F_k K_k H_k P_k^{-} F_k^T + Q_k \\
&= F_k P_k^{-} F_k^T - F_k P_k^{-} H_k^T (H_k P_k^{-} H_k^T + R_k)^{-1} H_k P_k^{-} F_k^T + Q_k
\end{aligned}$$

Discrete Riccati equation

- *a posteriori* Kalman filter

$$\begin{aligned}
\hat{x}_k^{+} &= (I - K_k H_k)(F_{k-1}\hat{x}_{k-1}^{+} + G_{k-1}u_{k-1}) + K_k y_k \\
P_k^{+} &= (I - K_k H_k)(F_{k-1}P_{k-1}^{+}F_{k-1}^T Q_{k-1})
\end{aligned}$$

Alternate Propagation of Covariance

Recall from Equation (5.19) the update equations for the estimation-error covariance:

$$\begin{aligned} P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\ P_k^+ &= P_k^- - P_k^-H_k^T(H_kP_k^-H_k^T + R_k)^{-1}H_kP_k^- \end{aligned} \tag{5.39}$$

If the $n \times n$ matrix P_k^- can be factored as

$$P_k^- = A_kB_k^{-1} \tag{5.40}$$

where A_k and B_k are $n \times n$ matrices to be determined, then P_{k+1}^- satisfies

$$P_{k+1}^- = A_{k+1}B_{k+1}^{-1} \tag{5.41}$$

where A and B are propagated as follows:

$$\begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} = \begin{bmatrix} (F_k + Q_kF_k^{-T}H_k^TR_k^{-1}H_k) & Q_kF_k^{-T} \\ F_k^{-T}H_k^TR_k^{-1}H_k & F_k^{-T} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \tag{5.42}$$

If F, Q, H, and R are constant matrices,

$$\begin{aligned} \begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} &= \begin{bmatrix} (F + QF^{-T}H^TR^{-1}H) & QF^{-T} \\ F^{-T}H^TR^{-1}H & F^{-T} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\ &= \Psi \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\ \begin{bmatrix} A_k \\ B_k \end{bmatrix} &= \Psi^{k-1} \begin{bmatrix} P_1^- \\ I \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} A_\infty \\ B_\infty \end{bmatrix} \approx \Psi^{2^p} \begin{bmatrix} P_1^- \\ I \end{bmatrix} \quad \text{for large } p \quad \Rightarrow \quad P_\infty^- = A_\infty B_\infty^{-1}$$

Steady-state covariance

Alternate Propagation of Covariance (Scalar Case with Constant F, Q, H, R)

$$
\begin{aligned}
\begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} &= \begin{bmatrix} F + \frac{H^2 Q}{FR} & \frac{Q}{F} \\ \frac{H^2}{FR} & \frac{1}{F} \end{bmatrix} \begin{bmatrix} A_k \\ B_k \end{bmatrix} \\
&= \Psi \begin{bmatrix} A_k \\ B_k \end{bmatrix}
\end{aligned}
$$

Eigenvalue decomposition of Psi

$$
\Psi = M \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} M^{-1}
$$

$$
\begin{aligned}
\begin{bmatrix} A_k \\ B_k \end{bmatrix} &= \Psi^{k-1} \begin{bmatrix} A_1 \\ B_1 \end{bmatrix} \\
&= M \begin{bmatrix} \lambda_1^{k-1} & 0 \\ 0 & \lambda_2^{k-1} \end{bmatrix} M^{-1} \begin{bmatrix} P_1^- \\ 1 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
P_k^- &= \frac{\tau_1 \mu_1^{k-1}(2RH^2 P_1^- - \tau_2) - \tau_2 \mu_2^{k-1}(2H^2 P_1^- - \tau_1)}{2H^2 \mu_1^{k-1}(2RH^2 P_1^- - \tau_2) - 2H^2 \mu_2^{k-1}(2H^2 P_1^- - \tau_1)} \\
\lambda_1 &= \frac{H^2 Q + R(F^2+1) + \sigma}{2FR} \\
\lambda_2 &= \frac{H^2 Q + R(F^2+1) - \sigma}{2FR} \\
\sigma &= \sqrt{H^2 Q + R(F+1)^2}\sqrt{H^2 Q + R(F-1)^2} \\
\tau_1 &= H^2 Q + R(F^2-1) + \sigma \\
\tau_2 &= H^2 Q + R(F^2-1) - \sigma \\
\mu_1 &= H^2 Q + R(F^2+1) + \sigma \\
\mu_2 &= H^2 Q + R(F^2+1) - \sigma \\
M &= \begin{bmatrix} \frac{\tau_1}{2H^2} & \frac{\tau_2}{2H^2} \\ 1 & 1 \end{bmatrix} \\
M^{-1} &= \frac{1}{\tau_1(R-1) + 2\sigma} \begin{bmatrix} 2RH^2 & -\tau_1 \\ -2RH^2 & R\tau_1 \end{bmatrix}
\end{aligned}
$$

$$
\boxed{\lim_{k\to\infty} P_k^- = \frac{\tau_1}{2H^2}}
$$

Example

$$
\begin{aligned}
x_{k+1} &= x_k + w_k \\
y_k &= x_k + v_k \\
w_k &\sim (0,1) \\
v_k &\sim (0,1)
\end{aligned}
$$

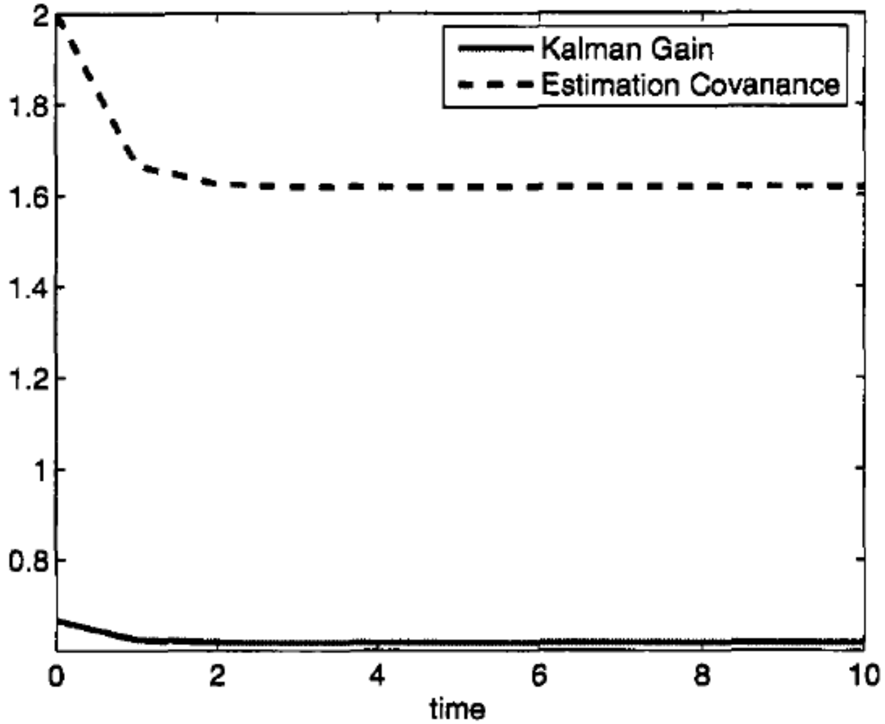
$$
\begin{aligned}
\tau_1 &= 1+\sqrt{5} \\
\tau_2 &= 1-\sqrt{5} \\
\mu_1 &= 3+\sqrt{5} \\
\mu_2 &= 3-\sqrt{5} \\
P_k^- &= \frac{\tau_1 \mu_1^{k-1}(2P_1^- - \tau_2) - \tau_2 \mu_2^{k-1}(2P_1^- - \tau_1)}{2\mu_1^{k-1}(2P_1^- - \tau_2) - 2\mu_2^{k-1}(2P_1^- - \tau_1)}
\end{aligned}
$$

$$
\begin{aligned}
P_\infty^- &= \frac{\tau_1}{2} \\
&= \frac{1+\sqrt{5}}{2} \\
&\approx 1.62
\end{aligned}
$$

$$
\begin{aligned}
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
&= \frac{P_k^-}{P_k^- + 1} \\
K_\infty &= \frac{1+\sqrt{5}}{3+\sqrt{5}} \\
&\approx 0.62
\end{aligned}
$$

$$
\begin{aligned}
P_\infty^+ &= \left(1 - \frac{1+\sqrt{5}}{3+\sqrt{5}}\right) \frac{1+\sqrt{5}}{2} \\
&= \frac{1+\sqrt{5}}{3+\sqrt{5}}
\end{aligned}
$$

$$P_k^+ = (I - K_k H_k) P_k^-$$



Divergence Issues

Practical Issues:

- Finite precision of a computer
- System model is not precisely known (F, Q, H, R, $\{w_k\}$, $\{v_k\}$)

Possible Solutions:

- Increase arithmetic precision
- Use some form of square root filtering
- Symmetrize P at each time step, e.g., $P = (P + P^T)/2$
- Initialize P appropriately to avoid large changes in P
- Use a fading-memory filter
- Use fictitious process noise

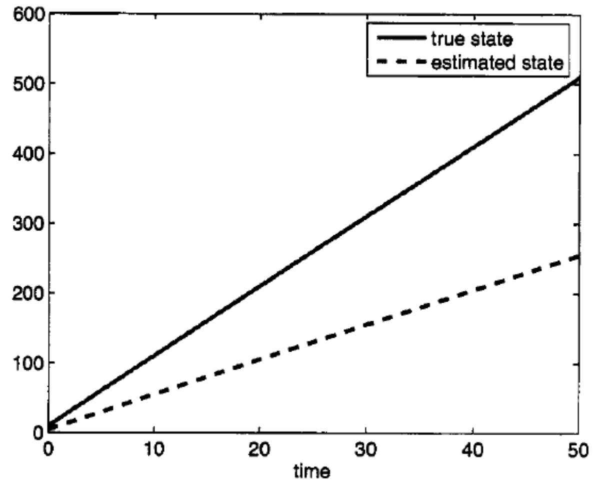
Example: Incorrect Modeling

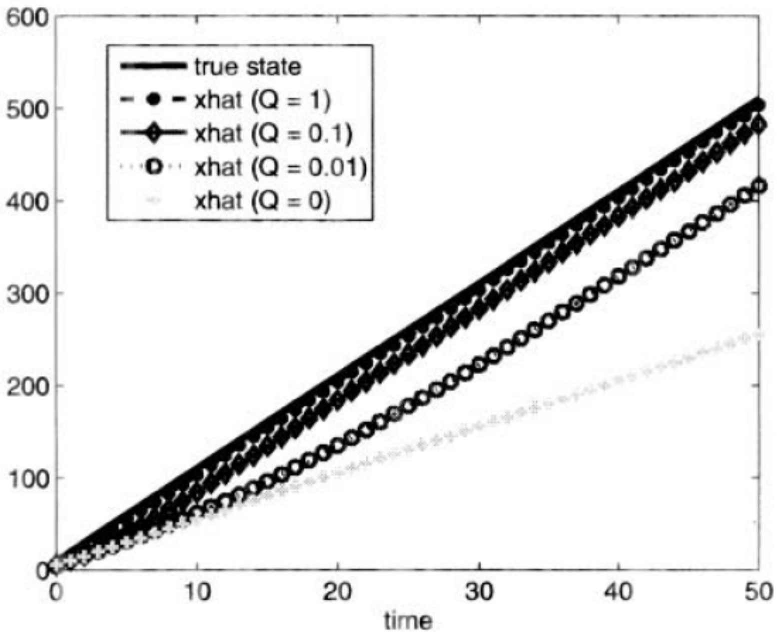
True System

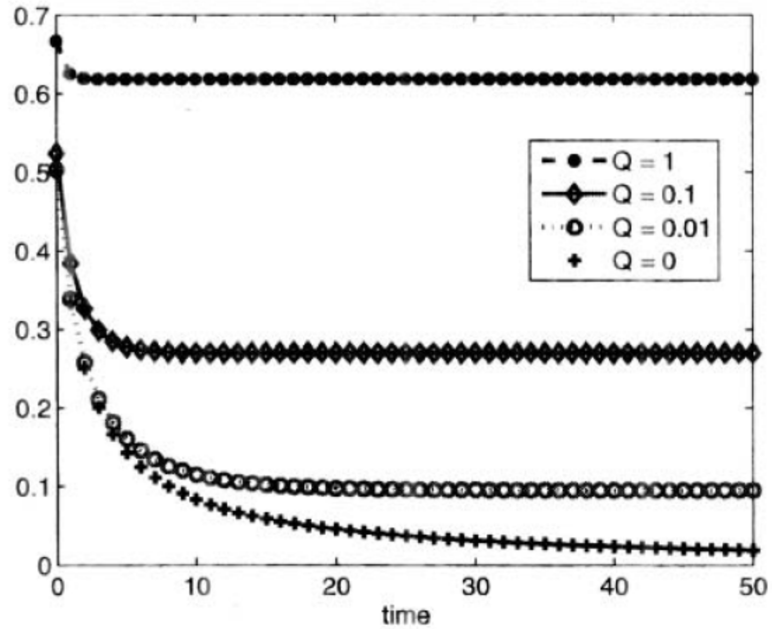
$$
\begin{aligned}
x_{1,k+1} &= x_{1,k} + x_{2,k} \\
x_{2,k+1} &= x_{2,k} \\
y_k &= x_{1,k} + v_k \\
v_k &\sim (0,1)
\end{aligned}
$$

Our Model

$$
\begin{aligned}
x_{k+1} &= x_k + w_k \\
y_k &= x_k + v_k \\
w_k &\sim (0,0) \\
v_k &\sim (0,1)
\end{aligned}
$$







$$P_{k+1}^- = F_k P_k^- F_k^T - F_k K_k H_k P_k^- F_k^T + Q_k$$

If Q ~ 0, then …

Kalman gain for different values of Q

Information Filter

Information matrix: $\boxed{\mathcal{I} := P^{-1}}$

- $\mathcal{I}$ represents the **certainty** in the state estimate.
- If $\mathcal{I} \to 0$, $P \to \infty$ and we have zero knowledge about x.
- If $\mathcal{I} \to \infty$, $P \to 0$ and we have perfect knowledge about x.

Matrix Inversion Lemma

$$
\begin{aligned}
(A + BC^{-1}D)^{-1} &= A^{-1} - A^{-1}B(C + DA^{-1}B)^{-1}DA^{-1} \\
(A + BC^{-1}D)^{-1}BC^{-1} &= A^{-1}B(C + DA^{-1}B)^{-1}
\end{aligned}
$$

$$
\begin{aligned}
P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\
\mathcal{I}_k^- &= \left(F_{k-1}(\mathcal{I}_{k-1}^+)^{-1}F_{k-1}^T + Q_{k-1}\right)^{-1} \\
&= Q_{k-1}^{-1} - Q_{k-1}^{-1}F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^TQ_{k-1}^{-1}F_{k-1})^{-1}F_{k-1}^TQ_{k-1}^{-1}
\end{aligned}
$$

$$
\begin{aligned}
P_k^+ &= \left((P_k^-)^{-1} + H_k^TR_k^{-1}H_k\right)^{-1} \\
(P_k^+)^{-1} &= (P_k^-)^{-1} + H_k^TR_k^{-1}H_k \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^TR_k^{-1}H_k
\end{aligned}
$$

Information Filter (1) $\mathcal{I} = P^{-1}$

$$
\begin{aligned}
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k \\
K_k &= (\mathcal{I}_k^+)^{-1} H_k^T R_k^{-1} \\
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-)
\end{aligned}
$$

Kalman Filter

$$
\begin{aligned}
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} \\
P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} (y_k - H_k \hat{x}_k^-) \\
P_k^+ &= P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-
\end{aligned}
$$

Information Filter Using the Canonical Parameterization

$$\begin{array}{rcl} \mathcal{I} & = & \Sigma^{-1} \\ z & = & \Sigma^{-1}\mu \end{array}$$

$$p(x|\mu,\Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

$$p(x|z,\mathcal{I}) = \exp\left(\alpha + z^T x - \frac{1}{2}x^T\mathcal{I}x\right)$$

$$\alpha = -\tfrac{1}{2}(n\log(2\pi) - \log|\mathcal{I}| + z^T\mathcal{I}z)$$

Define $\mathcal{I} = P^{-1}$ and $\hat{z} = P^{-1}\hat{x}$. Suppose $G_k = 0$.

$$\begin{aligned} \hat{z}_k^- &= (P_k^-)^{-1}\hat{x}_k^- \\ &= (P_k^-)^{-1}F_{k-1}\hat{x}_{k-1}^+ \\ &= (F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1})^{-1}F_{k-1}P_{k-1}^+\hat{z}_{k-1}^+ \\ &= Q_{k-1}^{-1}F_{k-1}\left((P_{k-1}^+)^{-1} + F_{k-1}^TQ_{k-1}^{-1}F_{k-1}\right)^{-1}\hat{z}_{k-1}^+ \\ &= Q_{k-1}^{-1}F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^TQ_{k-1}^{-1}F_{k-1})^{-1}\hat{z}_{k-1}^+ \end{aligned}$$

Matrix Inversion Lemma

$$\begin{aligned} (A+BC^{-1}D)^{-1} &= A^{-1} - A^{-1}B(C+DA^{-1}B)^{-1}DA^{-1} \\ (A+BC^{-1}D)^{-1}BC^{-1} &= A^{-1}B(C+DA^{-1}B)^{-1} \end{aligned}$$

$$
\begin{aligned}
\hat{z}_k^+ &= (P_k^+)^{-1}\hat{x}_k^+ \\
&= (P_k^+)^{-1}(\hat{x}_k^- + P_k^+ H_k^T R_k^{-1}(y_k - H_k\hat{x}_k^-)) \\
&= ((P_k^+)^{-1} - H_k^T R_k^{-1} H_k)\hat{x}_k^- + H_k^T R_k^{-1} y_k \\
&= ((P_k^+)^{-1} - H_k^T R_k^{-1} H_k)P_k^- \hat{z}_k^- + H_k^T R_k^{-1} y_k \\
&= \left((P_k^-)^{-1} + H_k^T R_k^{-1} H_k - H_k^T R_k^{-1} H_k\right) P_k^- \hat{z}_k^- + H_k^T R_k^{-1} y_k \\
&= \hat{z}_k^- + H_k^T R_k^{-1} y_k \\
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k
\end{aligned}
$$

Information Filter (2) $\mathcal{I} = P^{-1}$, $\hat{z} = P^{-1}\hat{x}$, and $G_k = 0$

$$
\begin{aligned}
\hat{z}_k^- &= Q_{k-1}^{-1} F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} \hat{z}_{k-1}^+ \\
\hat{z}_k^+ &= \hat{z}_k^- + H_k^T R_k^{-1} y_k \\
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k
\end{aligned}
$$

Kalman Filter

$$
\begin{aligned}
\hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \\
P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + P_k^-H_k^T(H_kP_k^-H_k^T + R_k)^{-1}(y_k - H_k\hat{x}_k^-) \\
P_k^+ &= P_k^- - P_k^-H_k^T(H_kP_k^-H_k^T + R_k)^{-1}H_kP_k^-
\end{aligned}
$$

r x r matrix inversion

Information Filter (2)

$\mathcal{I} = P^{-1}$, $\hat{z} = P^{-1}\hat{x}$, and $G_k = 0$

$$
\begin{aligned}
\hat{z}_k^- &= Q_{k-1}^{-1}F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^TQ_{k-1}^{-1}F_{k-1})^{-1}\hat{z}_{k-1}^+ \\
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1}F_{k-1}(\mathcal{I}_{k-1}^+ + F_{k-1}^TQ_{k-1}^{-1}F_{k-1})^{-1}F_{k-1}^TQ_{k-1}^{-1} \\
\hat{z}_k^+ &= \hat{z}_k^- + H_k^TR_k^{-1}y_k \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^TR_k^{-1}H_k
\end{aligned}
$$

n x n matrix inversion

- The information filter is more efficient than the Kalman filter if $n \ll r$ (more measurements than states).
- Still need to worry about the inversion of R_k.
- If $P_0^+ = \infty$, we can't run the Kalman filter but the information filter can be run with $\mathcal{I}_0^+ = 0$.
- If $\mathcal{I}_0^+ = \infty$, we can't run the information filter but the Kalman filter can be run with $P_0^+ = 0$.

Information Filter (1)

$\mathcal{I} = P^{-1}$

$$
\begin{aligned}
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k \\
K_k &= (\mathcal{I}_k^+)^{-1} H_k^T R_k^{-1} \\
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-)
\end{aligned}
$$

Information Filter (2)

$\mathcal{I} = P^{-1}$, $\hat{z} = P^{-1}\hat{x}$, and $G_k = 0$

$$
\begin{aligned}
\hat{z}_k^- &= Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} \hat{z}_{k-1}^+ \\
\mathcal{I}_k^- &= Q_{k-1}^{-1} - Q_{k-1}^{-1} F_{k-1} (\mathcal{I}_{k-1}^+ + F_{k-1}^T Q_{k-1}^{-1} F_{k-1})^{-1} F_{k-1}^T Q_{k-1}^{-1} \\
\hat{z}_k^+ &= \hat{z}_k^- + H_k^T R_k^{-1} y_k \\
\mathcal{I}_k^+ &= \mathcal{I}_k^- + H_k^T R_k^{-1} H_k
\end{aligned}
$$